Update
State of the Art Review of Space Radiation Monitors 2026

Heritage Systems, Miniaturisation and Emerging Cislunar Markets

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August 5, 2026

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	05/08/2026		07/10/2026

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Abstract

This report updates earlier surveys of space-trusted energetic-particle and radiation-monitoring instruments from the SWIMMR Project. It is written as an engineering reference. The review begins with the mature low-Earth-orbit (LEO), medium-Earth-orbit (MEO) and geostationary-orbit (GEO) markets, then considers highly elliptical, interplanetary and emerging cislunar applications. Instruments are compared using factual parameters such as detector technology, particle species, energy coverage, mass, power, volume, field of view, interfaces and flight heritage. Missing values are marked *ND*. The final section identifies technology gaps and emerging opportunities.

0.1 Purpose and Scope

The purpose of this survey is to provide engineers and mission designers with an updated, structured comparison of radiation monitors that have flown, are in development, or have recently been proposed. The review consolidates information from the SWIMMR S1 survey, the Kellett trusted-detector summaries, the 2020 radiation-monitoring presentation, and post-2023 literature and mission information. Information contained is limited to that which is publicly available in either peer review papers and organisational websites.

The document focuses on energetic electrons, protons, alpha particles and heavier ions; absorbed dose, dose rate, linear energy transfer (LET), charging and single-event-effect proxies; compact instruments suitable for hosted payloads and small spacecraft; and mission heritage from LEO to the 'new market' of lunar and cislunar space.

Energy coverage alone does not establish suitability. Geometric factor, shielding, saturation, contamination, dead time, angular response, calibration and data products remain decisive.

Scope Extension: Emerging Exploration Market In addition to well-established radiation monitors developed for Low Earth Orbit (LEO), Medium Earth Orbit (MEO) and Geostationary Earth Orbit (GEO) missions, this review also includes instruments designed for lunar surface, lunar orbital and cislunar applications. This reflects the rapid expansion of exploration activities beyond Earth orbit, driven by international (China) and commercial programmes such as Artemis, Gateway, Chang'E, Commercial Lunar Payload Services (CLPS), and an increasing number of robotic and commercial missions.

The coming decade is expected to see more lunar and cislunar missions than at any previous period in the Space Age. This rapid expansion is creating a substantial new market for compact radiation-monitoring instrumentation and is now one of the principal drivers of innovation in detector technology. [6, 7]

This is creating new requirements for compact, low-mass and low-power radiation monitors capable of operating in deep-space environments. As a consequence, a significant proportion of current detector innovation is now occurring within these emerging exploration programmes.

Although many of these instruments are being developed specifically for lunar and cislunar missions, they frequently represent the leading edge of radiation-monitor technology. Recent developments in hybrid silicon–scintillator detectors, Timepix-based imagers, compact spectrometers and multi-particle sensor systems have largely been driven by these new mission requirements. As these technologies mature, many are expected to transition into more traditional operational applications in LEO, MEO and GEO.

Accordingly, the inclusion of lunar and cislunar instruments in this review is intended not only to document the current state of the art for exploration missions, but also to identify technologies that are likely to influence the next generation of operational radiation monitors across all orbital regimes.

Chapter 1

Introduction: Why the Relevant Energy Range Changes with Mission Domain

1.0.1 Characterising the Space Enviroment

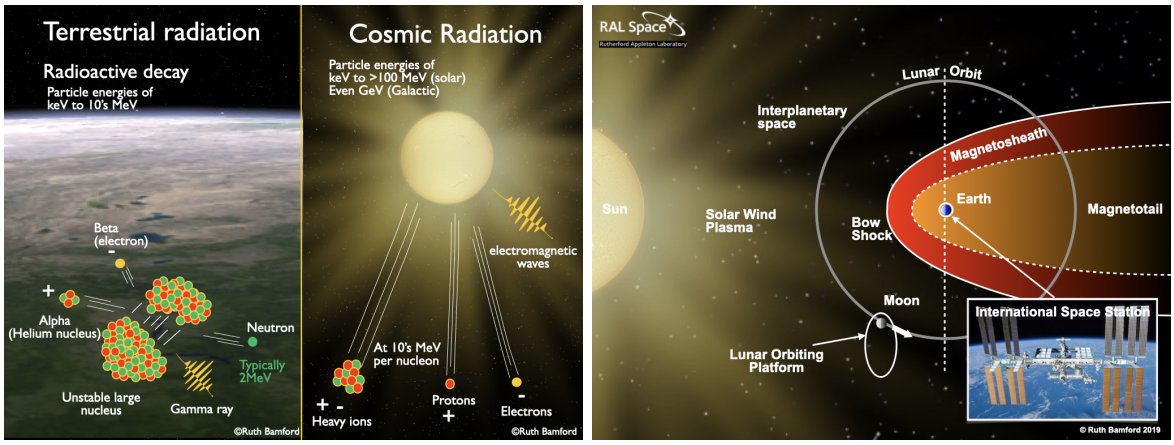


Figure 1.1: (left) The radiation in space is very different from type of radiation encountered from terrestrial source other than cosmic rays. (Right) The Moon orbits in and out of the Earth's magnetosphere and therefore has a complex and varied plasma environment.

The number and velocity of ions and electrons in space is changing all the time. The ions and electrons of space are quasi-neutral (a plasma) and originate from the Sun predominantly (Solar 'cosmic rays' SCR) but also violent events such as supernova and others from galactic sources and beyond(Galactic 'cosmic rays' GCR).

During periods of enhanced solar activity, such as solar flares and coronal mass ejections (CMEs), the flux of energetic charged particles can increase dramatically by many orders of magnitude, through Solar Energetic Particle (SEP) events. At even higher energies, Galactic Cosmic Rays (GCRs), believed to originate primarily from supernova remnants and other energetic astrophysical sources within our Galaxy, provide a persistent background radiation environment.

SEPs form a primarily concern to spacecraft operators (and astronauts) as they arrive suddenly without warning and contain highly penetrating ions and electrons. With CMEs there is some warning as it takes time for the plasma 'cloud' to reach Earth and it can be detected by monitoring spacecraft.

The Earth's magnetosphere changes the energy distribution of some of the ions and electrons but not all. Some are so energetic they pass straight through the magnetic boundaries. Others The become trapped within the Van Allen radiation belts.

The radiation of space is different in character to that encountered terrestrially other than cosmic rays. This is illustrated in Figure 1.1(far left and middle).

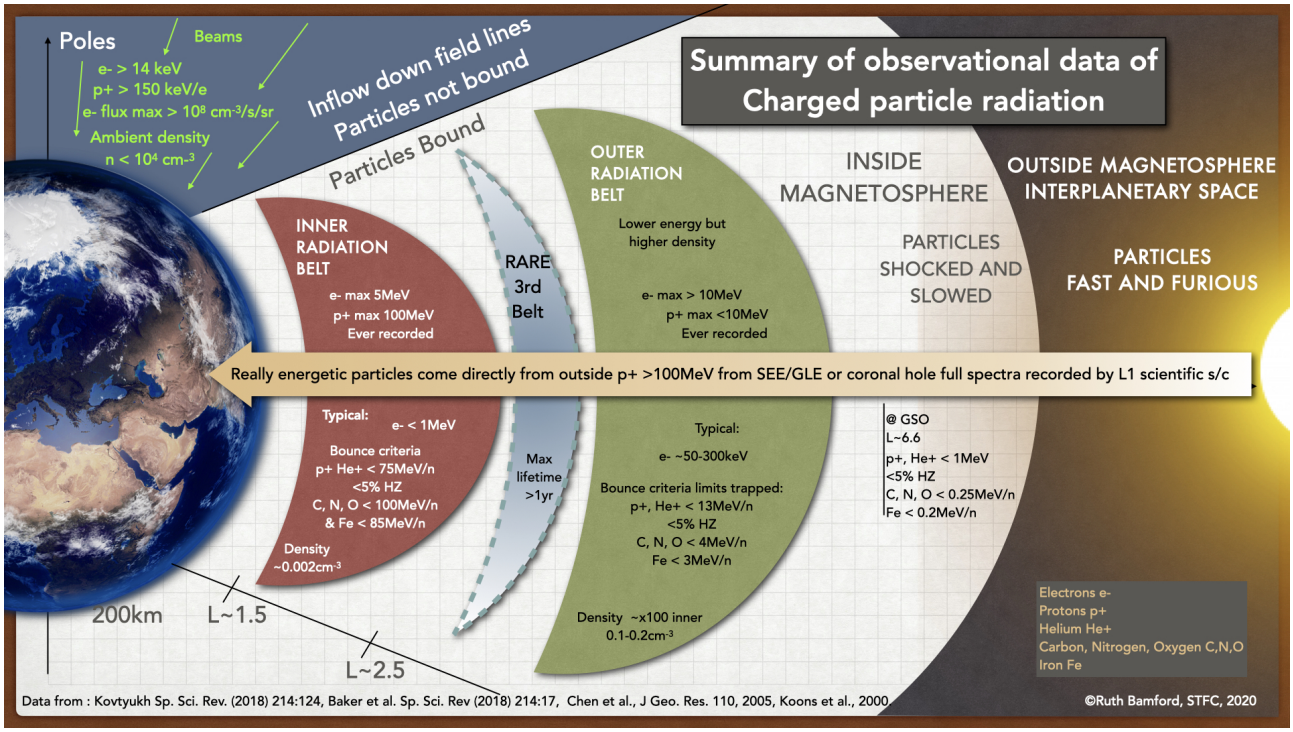


Figure 1.2: A graphically summary of the trapped and passing energy range of plasma particles expected within each region of the near Earth environment. Source data includes Kovtyukh Sp.Sci. Rev. (2018) 214:124, Baker et al. Sp. Sci. Rev (2018) 214:7, Chen et al., J. Geo. Res. 110, (2005) and Koons et. al., (2000). [3, 4, 5].

The Moon as emerging market The Moon orbits in and out of the Earth's magnetosphere and therefore has a complex and varied plasma environment as is shown in the far right hand panel Figure 1.1. For the majority of the orbit however spacecraft on or in orbit are outside the Earth's magnetosphere. Low cost sophisticated, multiple gravity assist orbits are what are opening the Moon as destination to commercial flyers.

The lunar environment is considerably more complex than either low Earth orbit or deep interplanetary space. During its 27.3-day orbit, the Moon periodically passes through the Earth's magnetotail before returning to the ambient solar wind. This is illustrated in the furthest right image in Figure 1.1. As a consequence, spacecraft operating in cislunar space or lunar orbit experience a changing radiation environment, alternating between geomagnetic influence and direct exposure to the interplanetary particle population. For the majority of each orbit, however, the Moon lies outside the Earth's protective magnetosphere and is exposed directly to solar energetic particles, galactic cosmic rays and locally generated secondary particles.

Table 1.1 summarises the principal radiation environments encountered by spacecraft together with the corresponding engineering measurement requirements. The near Earth environments are illustrated graphically in Figure 1.2.

1.1 Detector Technologies

The instruments reviewed in this survey employ a variety of physical detection principles. Nevertheless, they can be grouped into a number of broad detector families according to their underlying detection mechanism. Although individual instruments differ considerably in their implementation and performance, members of the same family generally share similar strengths, limitations and engineering trade-offs.

Table 1.1: Summary of radiation-monitor requirements by mission domain. Values are indicative rather than universal boundaries.

Domain	Principal populations	Energy concerns	Engineering consequence
LEO	Trapped protons, electrons, South Atlantic Anomaly, atmospheric secondaries	Electrons from keV to MeV; protons from MeV to hundreds of MeV	Large changes with altitude and inclination; high count rates in belt crossings; dose, displacement damage, charging and single-event effects.
MEO	Inner and outer radiation belts; persistent energetic electrons; trapped protons	Electrons from hundreds of keV to several MeV; protons from MeV to hundreds of MeV	Long-duration GNSS exposure requires stable calibration, wide dynamic range and charging measurements.
GEO	Outer-belt electrons, solar energetic particles, cosmic-ray background	Electrons from tens of keV to several MeV; SEP protons from MeV to hundreds of MeV and above	Operational monitoring requires continuous telemetry, directional response information and contamination control.
HEO	Repeated belt traversals plus solar-particle exposure	Broad electron and proton coverage, rapidly changing flux	High dynamic range and discrimination across very different environments within one orbit.
Interplanet	Solar wind, SEP, GCR and heavy ions	keV solar-wind plasma through GeV/nucleon GCR	Particle identification and high-energy response become increasingly important.
Cislunar	SEP, GCR, solar wind, magnetotail intervals, albedo particles and secondary neutrons/gamma rays	keV to GeV; charged and neutral secondaries	Compact multi-species monitors, directional measurements and neutron capability become more valuable.

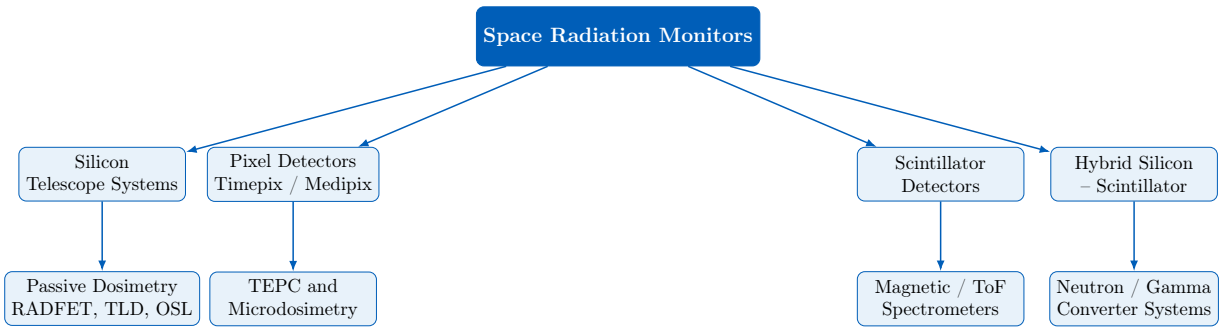


Figure 1.3: Principal detector technology families identified in this survey.

Table 1.2: Why different particle and energy channels are required.

Quantity	Indicative range	Primary engineering relevance	Typical measurement approach
Low-energy electrons	eV–100 keV	Surface charging; plasma environment	Electrostatic analyser or filtered solid-state detector
Energetic electrons	0.1–20 MeV	Internal charging, total dose, space-weather operations	Silicon telescope, scintillator or pixel detector
Protons	0.1–500 MeV and above	Dose, displacement damage, SEE, SEP monitoring	Silicon stack, scintillator telescope or hybrid calorimeter
Alpha particles	MeV/n to hundreds of MeV/n	SEP composition, dose and particle identification	Multi-layer ΔE–E telescope
Heavy ions	∼10 MeV/n to GeV/n	High-LET events and single-event effects	Silicon stack, magnetic spectrometer or calorimeter
Neutrons	Thermal to hundreds of MeV	Secondary radiation and dose equivalent	Converter plus scintillator or semiconductor
Gamma rays	keV–MeV and above	Secondary environment and spectroscopy	Scintillator or semiconductor spectrometer
Dose / LET	Mission-dependent	Crew exposure, electronics qualification, shielding validation	TEPC, silicon dosimeter, passive dosimeter or LET telescope
Charging current	fA–pA cm^{-2} scale	Internal dielectric charging and anomaly diagnosis	Shielded charge plates or dedicated charging monitor

Table 1.3: Engineering comparison of the principal detector technologies.

Technology	Strengths	Limitations	Typical Applications
Silicon telescope	High flight heritage; coincidence logic; ΔE–E particle identification; directional response	Detector thickness increases with required energy range; contamination corrections may be required	Operational radiation monitoring, space-weather monitoring and scientific particle measurements
Pixel detector	Event imaging; particle classification; directional reconstruction; compact and scalable	Limited stopping power; high-rate and pile-up constraints; calibration complexity	Miniaturised radiation monitors, distributed monitoring and track imaging
Scintillator	Large active volume; high stopping power for energetic particles; broad energy response	Particle identification often requires multiple detector materials or coincidence techniques	High-energy particle measurements, calorimetry, neutron and gamma-ray detection
Hybrid silicon–scintillator	Combines particle tracking with calorimetry; broad energy coverage	Greater mechanical, thermal and calibration complexity	Broad-energy charged-particle spectroscopy
TEPC	Direct tissue-equivalent microdosimetric response	Limited particle identification and directional information	Crew radiation monitoring and dose-equivalent estimation
Magnetic spectrometer / ToF	Charge sign, momentum and particle-species discrimination	Large mass, magnets, alignment tolerances and relatively high power	High-fidelity scientific instruments and emerging compact spectrometers
RADFET / Passive dosimeter	Very low power; compact; easily distributed throughout a spacecraft	Integrated dose only; little or no spectral information	Spacecraft housekeeping, shielding surveys and long-term dose monitoring

1.1.1 Silicon Telescope Systems

Representative instruments include CREDO, DosTel, CPDS, EMU, NGRM, ICARE-NG and REPT.

- Coincidence particle detection with ΔE–E capability (instrument dependent)
- Good proton and electron discrimination
- Typical mass: 1–4 kg
- Typical power: 3–10 W

Strength: High flight heritage, excellent stability and well-established calibration methods.
Limitation: Limited miniaturisation potential without performance trade-offs.

1.1.2 Pixel Detectors

Representative instruments include HMRM, SATRAM, LUCID, MIRAM, BIRD, HERA, MPT ISS and RISEPix.

- Event imaging and track reconstruction
- Particle classification and directional capability
- Typical mass: 50–300 g
- Typical power: <1–3 W

Strength: Excellent miniaturisation potential with highly scalable architectures.
Limitation: Limited stopping power and possible saturation during periods of high particle flux.

1.1.3 Hybrid Silicon–Scintillator Systems

Representative instruments include CRaTER, HEPD, GOES HEPAD, RAD and RADEM.

- Broad energy coverage
- Good charged-particle identification
- Heavy-ion capability
- Typical mass: 5–40 kg

Strength: Broad energy coverage with good particle-identification capability.
Limitation: Generally unsuitable for small hosted payloads because of mass and accommodation requirements.

1.1.4 Dosimetry and LET Monitors

Representative instruments include Liulin, M-42, RADFET-based systems, TEPC and DosTel.

- Integrated dose and/or dose-equivalent measurement
- High reliability
- Long operational heritage

Limitation: Limited spectral and temporal resolution.

1.1.5 Miniaturisation Trends

Radiation-monitor development has progressively moved towards lower mass, lower power and increased onboard processing capability. Broadly, three generations can be identified:

1. **Early Generation (1990–2010):** Classical silicon telescope instruments developed primarily for scientific missions.
2. **Operational Generation (2000–Present):** Compact engineering monitors for routine operation on LEO, MEO and GEO spacecraft.
3. **Miniaturised Generation (Present):** Pixel-based and hybrid detector systems intended for CubeSats, hosted payloads and emerging exploration missions.

Current pixel-based systems represent the most aggressive miniaturisation pathway. However, they do not yet provide the full energy coverage, stopping power or particle-identification capability of larger telescope systems.

1.1.6 Key Technical Trade-offs

The principal engineering trade-offs identified throughout this survey are:

1. Energy Range versus Instrument Size
2. Directional Response versus Omnidirectional Coverage
3. Dynamic Range versus Saturation
4. Mission-Specific Optimisation

Chapter 2

Engineering Comparison and Selection Matrices

Here we consider the metrics driving the community development pathways, explaining the development choices seen in literature.

Table 2.1: Indicative fit of detector technologies to mission domains. A tick means commonly suitable, not automatically qualified.

Technology	LEO	MEO	GEO	HEO	Interplanetary	Cislunar
Simple dosimeter / RADFET	✓	✓	✓	✓	✓	✓
Silicon telescope	✓	✓	✓	✓	✓	✓
Pixel detector	✓	✓	✓	✓	✓	✓
Si–scintillator hybrid	✓	✓	✓	✓	✓	✓
TEPC	✓	–	–	–	✓	✓
Magnetic spectrometer	–	–	–	✓	✓	✓
Neutron converter system	✓	–	–	–	✓	✓

Table 2.2: Questions to resolve before selecting or procuring a monitor.

Topic	Required question	Evidence required
Measurement objective	Operational warning, engineering housekeeping, anomaly diagnosis, science, dose or model validation?	User requirement and required data product
Species and energy	Which particle species and energy channels must be separated?	Response matrix and calibration report
Dynamic range	Can background and extreme events be measured without saturation?	Dead time, pile-up and maximum count-rate tests
Contamination	How are proton/electron cross-contamination and side-penetrating particles treated?	Monte Carlo response and beam calibration
Directionality	Is angular resolution required, or is an omnidirectional response preferable?	Geometric factor and angular-response model
Accommodation	Are mass, volume, peak power, thermal rejection and field of view compatible?	Mechanical, electrical and thermal ICD
Telemetry	Are raw events or processed products downlinked?	Data rate, onboard processing and storage budget
Radiation tolerance	Can the electronics survive the environment it measures?	TID, SEE and displacement-damage qualification
Heritage and TRL	Has the complete instrument flown, or only its detector component?	Flight history and configuration control
Calibration	Is calibration traceable across the complete energy range?	Beam, source and ground-calibration documentation

Chapter 3

Heritage and Existing Instruments

Table 3.1: Heritage and existing instruments extracted from the SWIMMR and Kellett surveys. *ND* denotes no data in the reviewed source.

Instrument	Detector technology	Protons	Electrons	Heavy ion / LET	Mass	Power	Heritage / domain
BIRD	Two Timepix units	*ND*	*ND*	*ND*	*ND*	Battery	Orion EFT-1; 5800 km apogee
CEASE / CEASE3	Multiple modules; silicon stacks and electrostatic analyser	2–150 MeV	100 eV–50 keV; 90 keV–5 MeV	*ND*	4 kg	8 W avg; 10 W peak	LEO, MEO, HEO, GEO
CREDO-II	Two 3 cm^2 silicon diodes in coincidence	*ND*	*ND*	*ND*	*ND*	*ND*	STRV missions
EMU	Silicon proton/electron monitor plus heavy-ion/TID functions	15–100 MeV	0.1–5 MeV	Yes	3.5 kg	8 W	Galileo MEO; Japanese GEO units
GOES SEM EPS/HEPAD	Solid-state pulse-height instruments	0.8–500 MeV; later channels > 700 MeV	0.6–4+ MeV	Alpha 4–500 MeV	*ND*	*ND*	Operational GEO
HERA	Network of Timepix sensor units	*ND*	*ND*	Track / dose classification	*ND*	*ND*	Orion / Artemis human-spaceflight monitor
HMRM	Miniaturised pixel silicon layers	0.6–500 MeV	0.035–6 MeV	dE/dx design aim	52 g	< 1 W	TechDemoSat-1; no useful returned data in 2022 review
LUCID	Five Timepix detectors on five faces	> 10 MeV	> 0.4 MeV	Alpha / other tracks	1.2 kg	5 W typical; 8 W max	635 km LEO, TDS-1
MERiT	APDs plus eight-SSD stack	0.2–0.4 and 7–100 MeV	0.005–0.2 and 1–8 MeV	*ND*	1.13 kg	0.3 W avg	3U CeRE; payload communications failure reported
MIRAM	Timepix3 plus four Si diodes	10(50)–400 MeV	0.5(3)–6.5 MeV	*ND*	> 150 g	> 1 W	Development concept; simulated ranges
MPT ISS	Two-sensor Modupix telescope	*ND*	*ND*	Direction and energy	*ND*	*ND*	ISS demonstration status uncertain
NGRM / RMU	Separate PSI electron and proton/ion sensor heads	2–200 MeV	0.1–7 MeV	LET spectra	1.4 kg	2 W; 6 W peak	EDRS-C, MTG, MetOp-SG, Sentinel-6 family
REM ISS	Timepix USB detectors, 300 and 500 μm sensors	*ND*	*ND*	Track morphology / dose	*ND*	*ND*	Seven units delivered to ISS
REPT	Multi-layer Micron silicon stack	17–200 MeV	1–20 MeV	*ND*	13.4 kg	6.2 W	Van Allen Probes, HEO
REPTile	Four Micron silicon detectors	8.5–40 MeV	0.5–> 3.3 MeV	*ND*	1.25 kg	2.5 W	CSSWE CubeSat, inclined LEO
RISEPix	Two stacked Timepix detectors	*ND*	*ND*	LET, particle type and direction	> 1 kg	> 1 W	RISESAT, likely LEO

Instrument	Detector technology	Protons	Electrons	Heavy ion / LET	Mass	Power	Heritage / domain
SATRAM	Single Timepix	*ND*	*ND*	Event topology	172–380 g	≤ 3 W	Proba-V, 820 km polar LEO
SEM-2	TED, MEPED and DPU suite	*ND*	*ND*	*ND*	18 kg total	*ND*	MetOp LEO
SREM	Three-detector ESA monitor	23–450 MeV	2–8 MeV	*ND*	2.6 kg	2.5 W	Eight ESA missions: LEO, HEO and interplanetary
TEPC ISS	A150 tissue-equivalent plastic and gas proportional counter	*ND*	*ND*	LET / dose equivalent	*ND*	*ND*	ISS crew-radiation monitoring
VZLUSAT-1	Single Timepix radiation monitor	*ND*	*ND*	Event topology	*ND*	*ND*	Czech nanosatellite, LEO

Chapter 4

Operational Markets and Mission Classes

Table 4.1: Representative instrument classes by market.

Market / mission class	Representative instruments	Typical priorities
LEO spacecraft health and space weather	SEM-2, SATRAM, HMRM, LUCID, RADMON, RISEPix, VZLUSAT-1	SAA response, trapped-particle flux, dose, anomaly correlation and low SWaP
MEO navigation	EMU, Merlin, CEDEX, SREM	Outer-belt electrons, long lifetime, internal charging and operational continuity
GEO meteorology and communications	GOES EPS/HEPAD, NGRM/RMU, SEDA, SEMIP, GRASP	Continuous operations, electron/proton channels, charging, cross-calibration and model validation
HEO / radiation-belt science	REPT, CREDANCE, CEASE3, NORM	Wide dynamic range, directional telescopes and repeated transitions between regimes
Human spaceflight	TEPC, ALTEA, DosTel, REM/Timepix, BIRD, HERA	Dose equivalent, shielding variations, real-time operations and distributed sensing
Interplanetary / planetary	SREM, MARIE, RAD, RADEM, RML	SEP and GCR response, autonomous operation, broad energy range and radiation tolerance
Emerging cislunar	CRaTER, RADOM, LND, MATCH, LunPAN concepts, IRADCAL	Primary and albedo particles, neutrons/gamma rays, directional context and limited heritage

Chapter 5

Developments Since 2023

The visible change since the original surveys is less a wholesale replacement of operational monitors and more an expansion toward compact, multi-function instruments for deep-space and cislunar markets.

Table 5.1: Recent and emerging instruments relevant to the updated market review. Status should be rechecked before procurement or mission selection.

Instrument	Mission / market	Particles / range	Technology	Update since 2023	Source status
MATCH	Chang'e-7 lunar orbiter; cislunar science and space-weather context	Cosmic and albedo particles; detailed range from payload paper	Symmetric DSSD and BGO scintillator stack	Payload design and ground-software work published in 2024–2025; delivery reported in 2025	Peer-reviewed payload/software papers and institutional mission update
Chang'e-7 energetic-particle detector	Chang'e-7 mission	Medium- and high-energy charged particles	Silicon detector and scintillator-based design	Detailed design paper published in *Aerospace*, 2024	DOI: 10.3390/aerospace11110893
LunPAN / PixPAN	ESA small-mission concept	Charged particles approximately 100 MeV–10 GeV in concept material	Compact magnetic spectrometer, Timepix4 tracking and ToF	Selected for ESA pre-Phase-A study; technical presentations published 2024–2025	Mission concept / conference material
LunPAN / NeuPix	ESA small-mission concept	Charged particles, neutrons and gamma rays; charged-particle band around 10–100 MeV	Pixel sensors with converter layers	Complementary neutral/charged-particle concept within LunPAN	Mission concept / conference material
IRADCAL	Turkish lunar mission / small-spacecraft concept	Electrons 50 keV–10 MeV; protons 0.4–150 MeV; heavy ions to about 20 MeV/n	Plastic and CsI(Tl) scintillators with SiPM readout	Concept paper published in 2024 for lunar albedo measurements	10.48550/arXiv.2411.11178
HENON REPE	Deep-space space-weather market; 12U CubeSat	Directional proton and electron fluxes	Miniaturised relativistic electron and proton telescope	Mission advanced in development; dedicated instrument paper published in 2026	ESA mission page and peer-reviewed instrument paper
Timepix4 concepts	Emerging small science and exploration payloads	Application-dependent	Data-driven pixel ASIC with time and energy information	Used in compact spectrometer concepts, particularly LunPAN/PixPAN	Component maturity exceeds integrated flight heritage

Caution. MATCH and the separately published Chang'e-7 energetic-particle detector may be distinct payload contributions or related detector activities. They should not be merged without confirmation from mission payload documentation.

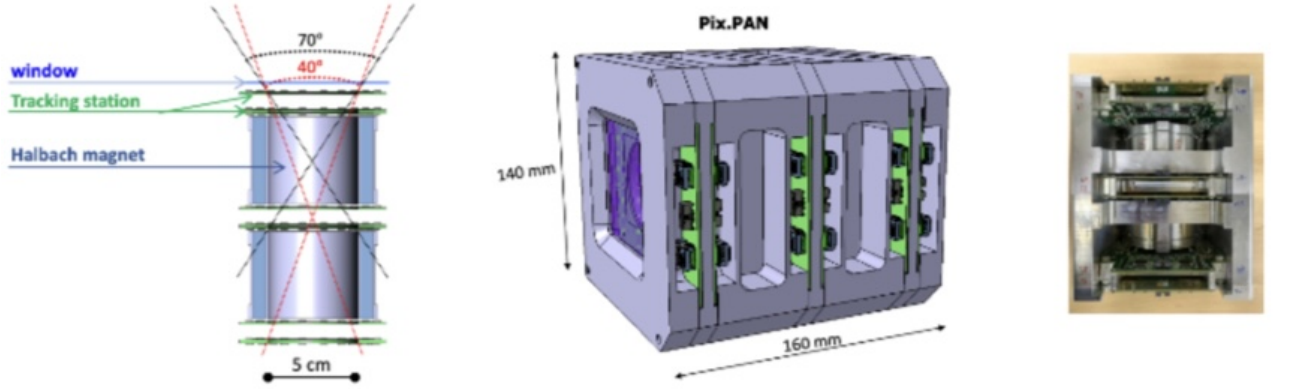


Figure 5.1: A sketch of the layout of Pix.PAN, showing the two magnet sectors and three tracking stations. (Middle) Preliminary design of the mechanical structure of the Pix.PAN spectrometer. (Right) The Mini.PAN spectrometer demonstrator. From [11]

Chapter 6

Technology Gaps and Emerging Opportunities

Table 6.1: Technology gaps and emerging opportunities.

Capability	Existing position	Gap	Opportunity
Broad-range low-SWaP spectrometry	Compact monitors generally cover selected bands or rely on unfolding	Simultaneous keV-to-hundreds-of-MeV coverage remains difficult below 1 kg	Hybrid silicon, scintillator and pixel architectures
High-rate SEP response	Several pixel systems provide rich event information	Pile-up, saturation and telemetry can dominate in extreme events	Data-driven ASICs, adaptive processing and dual-gain channels
Heavy-ion identification	Available in larger stacks and spectrometers	Limited capability in very small monitors	Multi-layer pixels, ToF and compact magnetic analysis
Neutron measurement	Mature for resource mapping and large dosimeters	Charged and neutral radiation are often separate payloads	Converter-coated pixels and segmented scintillators
Directional measurements	Telescopes provide direction but reduce acceptance	Small omnidirectional monitors may not distinguish incident and albedo populations	Multiple sensor heads, stacked pixels and compact hodoscopes
Cislunar heritage	LRO, Chang'e and Chandrayaan provide important data	Few operational monitors are qualified for distributed cislunar infrastructure	Hosted payloads on logistics vehicles, relay orbiters, landers and surface assets
Distributed monitoring	ISS and some spacecraft use multiple dose sensors	Few networks provide cross-calibrated spectra across platforms	Common compact monitor, calibration and interoperable data products
Standardised data products	Operational systems use mission-specific processing	Cross-mission comparison remains difficult	Open response matrices and agreed calibration metadata
UK sovereign supply	UK heritage exists in silicon diodes, electronics and instrument engineering	Several leading pixel solutions depend on non-UK ASIC supply chains	UK integration, calibration, packaging and mixed-technology instruments

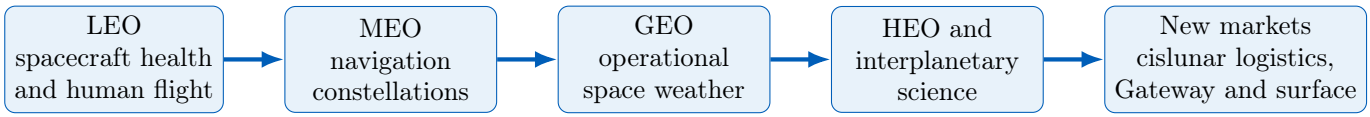


Figure 6.1: Market progression from established Earth-orbit services to emerging cislunar infrastructure.

Chapter 7

Conclusions

The established radiation-monitor market remains concentrated in LEO, MEO and GEO, where operational requirements are comparatively mature. Silicon telescopes and multi-sensor suites remain the most defensible solution when species separation, wide energy coverage and calibration fidelity are required. Pixel detectors have expanded the low-SWaP design space, particularly for event imaging, distributed monitoring and small spacecraft, but they do not automatically replace thicker telescope or calorimeter architectures.

The emerging cislunar market changes the balance of requirements rather than the underlying detector physics. Instruments must operate outside continuous geomagnetic shielding, may need to distinguish incident from albedo populations, and increasingly combine charged-particle, neutron and gamma-ray measurements within restricted accommodation budgets. The strongest opportunity here being pursued is not a single universal sensor, but a modular, cross-calibrated family spanning operational dosimetry, compact spectrometry and higher-fidelity science measurements. However missions like ARGONAUT and NovaMoon and similar offer opportunities to be on service providing platforms for those exploring the Moon.

Appendix A

Appendix

A.1 Orbit-Grouped Instrument Tables

The tables in this appendix provide alternative views for readers beginning with a mission domain. They are indexes into the canonical heritage and post-2023 tables (Tables 3.1 and 5.1); quantitative values should be maintained in those tables rather than independently here.

Table A.1: Orbit-grouped instrument index.

Domain	Representative instruments	Principal populations	Selection questions
LEO	HMRM, SATRAM, LUCID, HEPD, DosTel, ALTEA	Trapped particles, SEP and GCR	SAA cadence, dynamic range, platform size and human-rating requirements
MEO	SREM, NGRM, EMU, Merlin and CEDEX	Inner- and outer-belt populations	Dose tolerance, electron/proton discrimination and long-term stability
GEO	SREM, NGRM/RMU, GOES particle suites and SEDA	Outer-belt electrons, SEP and GCR	Latency, charging relevance, contamination and cross-calibration
HEO	REPT, CREDANCE, CEASE3 and NORM	Belt crossings, SEP and GCR	Changing geometry, background and autonomous mode control
Interplanetary	RAD, RADEM, Liulin family and HENON REPE	SEP, GCR and solar-wind particles	Species, directionality, event response and autonomous operation
Cislunar/lunar	CRaTER, LND, MATCH, PixPAN, NeuPix and IRADCAL	SEP, GCR and lunar secondaries	Albedo directionality, neutral particles, resources and limited heritage

A.2 Human-Exploration Detector Table

Table A.2: Detector-family index and technology colour key.

Family	Representative instruments	Comparison fields to retain
Silicon telescope	SREM, NGRM, EMU, ALTEA and HENON REPE	Stack geometry, coincidence, geometric factor and contamination
Pixel detector	HMRM, SATRAM, LUCID, REM and RISEPix	Sensor thickness, pitch, acquisition mode, pile-up and classification
Scintillator	Instrument-specific counters and calorimeters	Material, light collection, threshold and radiation ageing
Hybrid	REPT, CRaTER, RAD, HEPD, MATCH and IRADCAL	Tracker/calorimeter coupling, stopping range and cross-calibration
Dosimetry/TEPC	TEPC, DosTel and Liulin family	Dose quantity, tissue equivalence, LET and integration period
Neutral converter	NeuPix and dedicated neutron/gamma systems	Converter, efficiency, background rejection and unfolding
Magnetic/ToF	PixPAN and larger spectrometers	Field integral, lever arm, ToF resolution and charge-sign determination

Table A.3: Human-exploration detector view.

Instrument/fami	Technology	Environment/herita	Human-exploration relevance
ALTEA	Silicon telescope array	ISS	Directional and LET-related measurements; configuration and status to verify
TEPC	Tissue-equivalent proportional counter	Crewed-spacecraft heritage	Direct microdosimetric relevance with limited species resolution
DosTel	Silicon dosimeter/telescope	ISS	Operational absorbed-dose and particle information
REM/Timepix	Pixel detector	ISS	Distributed track imaging and dose mapping
Liulin family	Silicon dosimeter	Multiple crewed and deep-space missions	Compact long-duration dose monitoring
CRaTER	Silicon/tissue-equivalent-plastic stack	Lunar orbit	Shielding and tissue-equivalent-material response context
RAD	Silicon/scintillator	Cruise and Mars surface	Mixed-field dose and particle-environment context

Source Documents and References

Bibliography

[1] B. J. Kellett, *SWIMMR S1 Space Trusted Existing Sensors Survey*, Document SWMR-S1ST-RAL-RS-0001, Issue 0.9, STFC, 30 June 2020.

[2] B. J. Kellett, *Summary of Trusted Detectors for Space*, Updated Version 2, January 2022.

[3] A. S. Kovtyukh, "Space Radiation Environment Monitoring: Current Status and Future Prospects," *Space Science Reviews*, Vol. 214, Article 124, 2018. doi:10.1007/s11214-018-0553-y

[4] Baker, D. N., Erickson, P. J., Fennell, J. F., Foster, J. C., Jaynes, A. N. and Verronen, P. T., "Space Weather Effects in the Earth's Radiation Belts," *Space Science Reviews*, Vol. 214, Article 7, 2018. doi:10.1007/s11214-017-0452-7

[5] Koons, H. C., Fennell, J. F., Chen, M. W. and Blake, J. B., "Internal Charging: A Preliminary Environmental Specification for Satellites," *IEEE Transactions on Plasma Science*, Vol. 28, No. 6, pp. 2029–2036, 2000. doi:10.1109/27.902230

[6] NASA, "Strategic Plans for Lunar Missions," Planetary Advisory Committee Presentation, November 2023. Available at: https://science.nasa.gov/wp-content/uploads/2024/08/day2-1630-pac-lunarstrategy-noble-nov2023.pdf

[7] Swope, B., "Swimming Upstream: The Future of Cislunar Activity," Center for Strategic and International Studies (CSIS), October 2024. https://aerospace.csis.org/wp-content/uploads/2024/10/241021_Swope_Swimmimg_Upstream_0-compressed.pdf

[8] *Radiation Monitoring Packages*, presentation, March 2020.

[9] L. Wang et al., "Design and Development of Energy Particle Detector on China's Chang'e-7," *Aerospace*, 11(11), 893, 2024. doi:10.3390/aerospace11110893.

[10] T. Panyalert et al., "Ground software architecture for a lunar particle detector," *Advances in Space Research*, 2025. Publisher page.

[11] X. Wu, "LunPAN: Comprehensive Study of the Lunar Energetic Particle Environment," Lunar Surface Science Workshop, 2024. Conference paper.

[12] A. B. Alpat et al., "IRADCAL," 2024. doi:10.48550/arXiv.2411.11178.

[13] European Space Agency, "HENON." ESA mission page.

[14] C. Ngom et al., "Relativistic Electron and Proton Experiment for the HENON mission," *Journal of Space Weather and Space Climate*, 2026. Journal article.

[15] Z. Xu et al., "Primary and albedo protons detected by the Lunar Lander Neutrons and Dosimetry experiment," *Frontiers in Astronomy and Space Sciences*, 2022. doi:10.3389/fspas.2022.974946.

Appendix A

Detector Review Presentation

RAL Space

Space Radiation Monitors

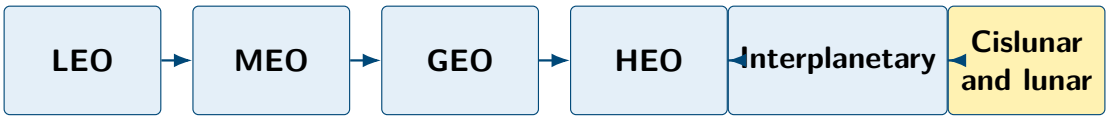
Heritage systems, emerging technologies and engineering gaps

Dr Ruth Bamford | RAL Space

State-of-the-art review update, 2026

The review connects mature markets to emerging missions

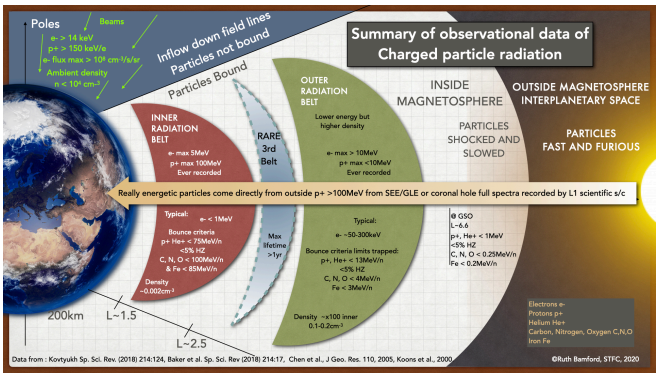
The survey is organised around the engineering question: **what measurement capability fits the mission?**



- The Kellett/SWIMMR material provides the heritage baseline.
- Post-2023 developments are compared separately, using the same engineering fields.
- Cislunar systems are included as an emerging market and a source of detector innovation.

Mission domain changes the useful energy range

- **LEO:** trapped particles, SAA passages and atmospheric secondaries.
- **MEO/GEO:** persistent energetic electrons, charging and solar-particle exposure.
- **Beyond Earth:** SEP, GCR, heavy ions and locally generated secondaries.



Engineering consequence:
No single energy range or detector architecture is optimal across every domain.

Detector families occupy different parts of the trade space

Family	Principal strength	Principal limitation
Silicon telescope	Heritage and ΔE–E identification	Thickness, geometry and contamination
Pixel / Timepix	Imaging, track classification and low SWaP	Stopping power and high-rate response
Scintillator	Active volume and high-energy stopping power	Species separation needs added information
Hybrid Si–scintillator	Tracking plus calorimetry over a broad range	Integration and cross-calibration complexity
Dosimetry / TEPC	Dose relevance and distributed monitoring	Limited spectral or directional information
Magnetic / ToF	Momentum, charge sign and composition	Mass, power and alignment

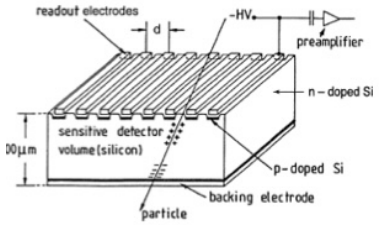
Heritage remains valuable ĂŤbut the instruments are not interchangeable

Operational heritage

Silicon telescopes and multi-sensor suites provide defensible calibration, species separation and wide energy coverage.

- SREM: multi-mission Earth-orbit and interplanetary heritage.
- NGRM/RMU: compact operational electron and proton/ion sensor heads.
- REPT: high-fidelity radiation-belt science.

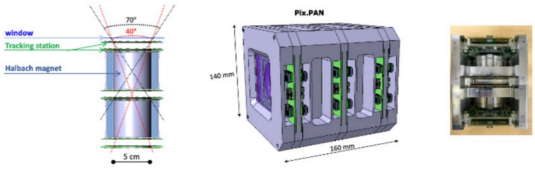
Solid-State Detectors (SSDs).jpeg



Miniaturised heritage

Timepix-based instruments add track imaging and distributed sensing, with different limits on stopping power and flux.

Post-2023 work expands the architecture, not just the catalogue



Pix.PAN combines magnetic analysis, Timepix4 tracking and time-of-flight.

- **MATCH:** DSSD and BGO stack for lunar particles.
- **PixPAN / NeuPix:** complementary charged- and neutral-particle concepts.
- **IRADCAL:** compact scintillator/SiPM system for lunar albedo studies.
- **HENON REPE:** directional electrons and protons from a 12U platform.

Common direction: more particle information within constrained accommodation.

The remaining gaps are system-level gaps

Capability	Present limitation	Opportunity
Broad-range low-SWaP	Selected bands or model-dependent unfolding	Hybrid and stacked architectures
Extreme SEP response	Pile-up, saturation and telemetry demand	Adaptive processing and dual gain
Compact neutron measurement	Often a separate payload	Converter pixels and segmented scintillators
Directional measurements	Acceptance competes with angular information	Stacked pixels and multiple sensor heads
Cross-mission comparison	Mission-specific response and products	Open response matrices and calibration metadata

The priority is not a universal sensor; it is a **cross-calibrated family of instruments**.

Select the measurement before selecting the detector

1. Define the operational or scientific data product.
2. Specify species, energy, cadence and dynamic range.
3. Test contamination, saturation and angular response.
4. Confirm mass, power, telemetry, interfaces and heritage.
5. Require a traceable calibration and response matrix.

Review conclusion

Pixel and hybrid systems broaden the available trade space, but they complement rather than automatically replace mature telescopes, calorimeters and dosimeters.

Next step

Complete the controlled comparison dataset and verify quantitative fields against primary sources.

Previous Presentation Tables

Orbit-Based View of Detector Developments

- Instruments are grouped by measurement environment:
  - heritage LEO mapping / dosimetry systems
  - MEO radiation-belt instruments
  - cislunar / lunar-environment concepts
- Highlighting used in the tables:
  - Red: not in the original baseline list
  - Blue: baseline instrument with updated / changed performance or status

LEO Instruments

Instrument	Type	Proton (MeV)	Mass	Status
HMRM	Pixel Si	0.6–500	52 g	flown
LUCID	Timepix array	>10	1.2 kg	flown
SATRAM	Timepix	ND	172–380 g	flown
MIRAM	Timepix3 + Si	10–400	>150 g	proposed / development
MERiT	SSD + APD	7–100	1.13 kg	flown
RADMON (Aalto-1)	Si + CsI	10–80	360 g	flown
RISEPix	2× Timepix stack	ND	>1 kg	flown
VZLUSAT-1	Timepix	ND	ND	flown
ALTEA	Si telescopes	ND	ND	flown
DosTel	Si telescope	ND	ND	flown
TEPC	Gas proportional counter	ND	ND	flown
HEPD	Si + scintillator	30–200	43 kg	flown

LEO remains dominated by miniaturised, mapping, and dosimetry-oriented systems, especially Timepix-class detectors.

MEO Instruments

Instrument	Type	Proton (MeV)	Mass	Status
EMU	Si telescope	15–100	3.5 kg	flown
Merlin	Si + SURF	>40	1.7 kg	flown
CEDEX	Si-based	ND	ND	flown
SREM (GIOVE-B)	Multi-detector	23–450	2.6 kg	flown
NGRM	Si (PSI)	2–200	1.4 kg	flown
ICARE-NG	Si coincidence	27–292	2.8 kg	flown
CEASE3	Si stacks	2–150	4 kg	flown
CREDANCE	Si detectors	ND	ND	flown
REPT	Si stack	17–200	13.4 kg	flown

MEO systems are primarily radiation-belt instruments with thicker silicon stacks and stronger spectral capability.

Cislunar / Lunar-Orbit Instruments

Instrument	Type	Energy Range	Platform	Status
CRaTER	Si pairs + tissue-equivalent plastic	ND	LRO lunar orbit	flown
RADOM	Mini dosimeter spectrometer	ND	Chandrayaan-1 lunar orbit	flown
MATCH	DSSD + BGO	MeV range	Chang'E-7	development
PixPAN	Magnetic spectrometer + Timepix4	0.1–10 GeV	ESA LunPAN	concept
NeuPix	Hybrid pixel detector	10–100 MeV	ESA LunPAN	concept
IRADCAL	Scintillator + SiPM	keV–100 MeV	2U CubeSat concept	concept

Cislunar developments are increasingly hybrid, with explicit focus on lunar albedo particles, neutrons and gamma-rays.

Comparison Across Orbit Regimes

Orbit Regime	Dominant Detector Style	Main Driver	Typ
LEO	Pixel / compact Si / dosimetry	mapping, dose, variability	lo
MEO	Silicon telescope stacks	trapped belts / spectra	high
Cislunar	Hybrid pixel + scintillator + spectrometer	SEP, GCR, albedo, neutrons	multi-sp

- LEO: compact deployment and distributed monitoring
- MEO: spectral discrimination in radiation belts
- Cislunar: broad environment, secondaries from regolith, neutron sensitivity

Physics Implications

- **LEO:** compact instruments are often sufficient for dose monitoring and spatial mapping
- **MEO:** thicker detector stacks are required to resolve trapped particle populations
- **Cislunar:** instrument requirements broaden to include:
  - SEP protons in the 10–100 MeV range
  - GCR response at higher energies
  - secondary neutrons and gamma-rays from regolith
- This is why recent cislunar concepts are increasingly hybrid rather than pure silicon telescope or pure pixel systems

Summary

- Grouping instruments by orbit clarifies both technology choice and physics driver
- Red entries identify instruments absent from the original baseline list
- Blue entries identify baseline instruments whose status / performance has changed
- LEO remains dominated by compact and pixel-based monitors
- MEO remains the domain of silicon telescope systems
- Cislunar developments show the clearest shift toward hybrid architectures